Nested Classes

- Static Nested
- Inner class
- Local class
- Anonymous class

```
class A {
    class B {
    }
}
```

① Logical Grouping.

```
class A {
}

class B {
}

class C {
}
```

② Better access to Outer class.

```
class Outer {
    private int x;
    class Inner {
    }
}
```

Static Nested Class

```
class Outer {
    static class Inner {
    }
}
```

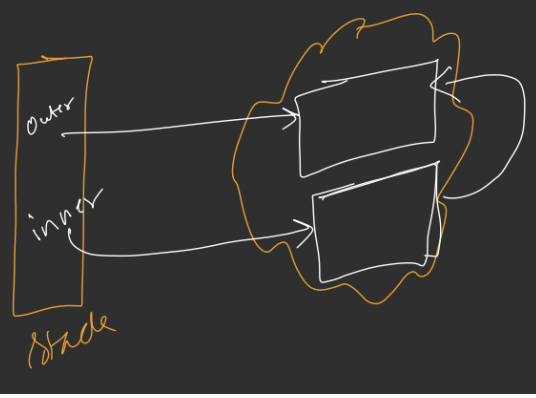
Key properties

1. Does not need an instance of Outer class.
2. Class can be instantiated like Normal class.
3. Can access only static members of outer class.
4. Can access non static members by having a reference of Outer class.
5. It is just like Normal class & can do anything an Outer class does.

Inner Class

```
class Outer {
    class Inner {
        Outer outer;
    }
}
```

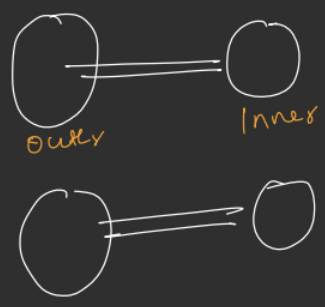
```
Outer outer = new Outer();
Outer.Inner inner = Outer.new Inner();
```



```
class Outer {
    class Inner {
    }
}
```

Static Member

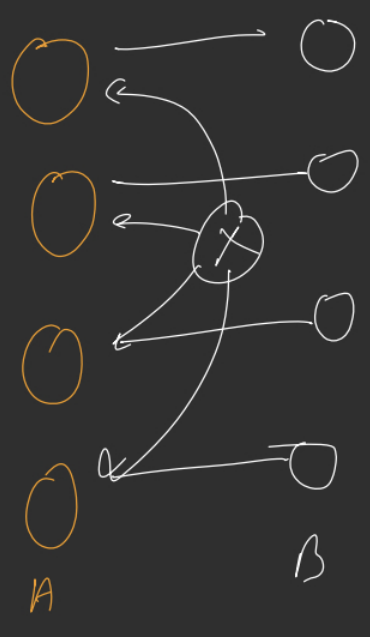
Non-static "




```
class Outer {
    int x = 10;
    class Inner {
        int x = 20;
        ____(x);      // 20
        ____(Outer.this.x);
    }
}
```

Inner class → Static Members (X)

```
class A {
    class B {
        static int x;
    }
}
```

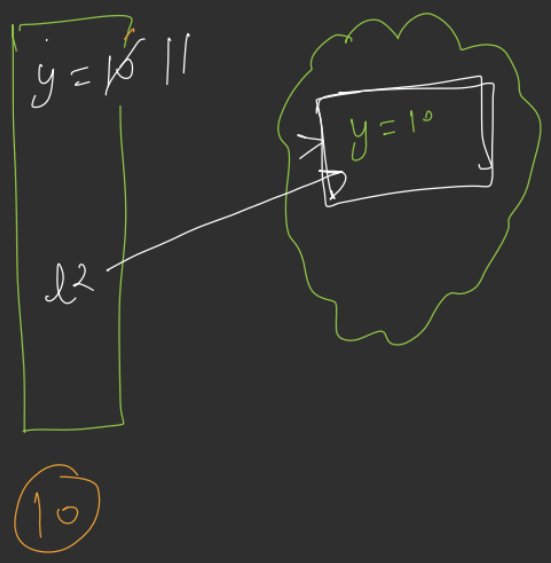


Local class

→ if() {}, for() {}

→ fun() {—}

```
{
    class local {
    }
}
```

```
void fun() {
    class local {
    }
}
```

```
class Outer {
    Object greet() {
        int y = 10;        y++;
        class Local {
            int y = 10;
            void fun() {
                system.out.println(y);
            }
        }
        Local l = new Local();
        l.fun();
        return l;
    }
    void fun() {
        Object o = greet();
    }
}
```